ASTRACT ALGEBRA

HOMEWORK 2

Please hand in your solutions to the following problems by 2025-07-18. The homework has to be submitted via the course website as a single pdf file. It has to be typeset; handwritten homework will not be graded as the platform won't accept it. TeX is strongly preferred, but we will accept homework typeset with MS Word or similar, so long as the text is easy to read and meets reasonable standards of mathematical writing. You are encouraged to discuss the problems with your colleagues, but must hand in your own solutions. You are welcome to ask us questions, especially in connection to the discussion session.

Problem 1. (2 points)
Let P be a p-Sylow of a finite group G. Show that if N is a normal subgroup of G, then $N \cap P$ is a p-Sylow of N. Give an explicite p-Sylow of G/N.

Problem 2. (3 points)
Let $f : G \to H$ be a group homomorphism, where G is finite. Denote N its kernel.

(1) Let P and Q be two p-Sylow of G. Show that if $f(P) = f(Q)$ then P and Q are conjugate in G. **(1 point)**
(2) Assume that H is also finite and f is surjective. Let P be a p-Sylow of G and let $N_G(P)$ be its normaliser in G. Show that $f(N_G(P)) = N_H(f(P))$. **(1 point)**
(3) Assume that p does not divide the order of H. Then $f(P) \subset \ker(f)$ and $f(N_G(P)) = H$. **(1 point)**

Problem 3. (8 points)

(1) Let G be a group of order 15. Show that:
  (a) G has a normal subgroup. **(1 point)**
  (b) G is cyclic. **(2 point)**
(2) Let G be a finite group of order pq where p and q are prime numbers with $p > q$. Assume $p \not\equiv 1 \pmod{q}$. Show that G is cyclic. **(2 point)**
(3) Let G be a group of order 12.
  (a) Show that G has a normal subgroup. **(2 point)**
  (b) Give a non cyclic group of order 12. **(1 point)**

Problem 4. (2 points)
Let $a = (1234)(5678),\ b = (1537)(2648)$ be elements of S_8. Show that $H = \langle a, b\rangle$ is an abelian group of order 8 and decompose it into a product of cyclic groups.

Problem 5. (3 points)

(1) Let G be a group, and let Z be a central subgroup of G (that is, Z is contained in the center of G). Show that if G/Z is cyclic then G is abelian. **(1 point)**
(2) Let G be group of order p^2. Show that G is abelian. **(1 point)**
(3) Give a list of all groups of order p^2 up to isomorphism. **(1 point)**

Problem 6. (2 points)
Let A be an abelian group. The dual of A is the set $\hat{A} := \mathrm{Hom}(A, \mathbb{C})$ endowed with the usual composition.

(1) Show that if A is cyclic of order n so is $\hat{A}$. **(1 point)**
(2) Show that if A is abelian of order n so is $\hat{A}$. **(1 point)**